Quiz 1: Contradiction and Induction

Claude

60 minutes. Closed book, no notes, no computer. Each problem is worth 10 points.

Problem 1. Prove that $3^{2n+1} + 2^{n+2}$ is divisible by 7 for every integer $n \geq 0$.

Problem 2. The sequence $a_1, a_2, \ldots$ is defined by $a_1 = 1$ and $a_{n+1} = a_n + \dfrac{1}{a_n}$ for $n \geq 1$. Prove that $14 < a_{100} < 18$.

Problem 3. Let n be a positive integer. The numbers $1, 2, \ldots, 2n$ are split into two groups of n numbers each. The numbers in the first group are written in increasing order, $a_1 < a_2 < \cdots < a_n$, and the numbers in the second group in decreasing order, $b_1 > b_2 > \cdots > b_n$. Prove that

$$|a_1 - b_1| + |a_2 - b_2| + \cdots + |a_n - b_n| = n^2.$$